

Age Classification using a Strongly Regularized ResNet-18

A ResNet-18 architecture was again employed as the primary model while strictly adhering to the assignment constraint that the network must be trained from scratch (`weights=None`). The core backbone architecture remained unchanged to maintain compliance with the provided model restrictions. Only the classification head was modified to introduce additional regularization and improved feature stabilization. Specifically, a **Batch Normalization layer** was added before the fully connected projection, followed by **Dropout (0.4)** and a two-layer linear projection ($512 \rightarrow 256 \rightarrow 2$). This modification stabilizes the feature distribution produced by the backbone and reduces overfitting by encouraging distributed feature representations.

Enhanced Data Augmentation and Regularization: To further improve robustness and prevent memorization of the training set, a stronger augmentation pipeline was introduced. Training images were processed using **Random Resized Cropping**, **Horizontal Flipping**, and **RandAugment**, which applies stochastic geometric and photometric transformations. Additionally, **Random Erasing** was applied to randomly mask image regions so that the network learns global facial structure rather than relying on localized features.

Further regularization was achieved through **MixUp augmentation**, where pairs of training samples are linearly combined to produce synthetic inputs with interpolated labels. This technique encourages smoother decision boundaries and improves generalization under limited data conditions. In addition, **Label Smoothing** was applied within the cross-entropy loss to prevent overly confident predictions during training.

Model Stabilization through Exponential Moving Average: To improve the stability of the learned parameters, an **Exponential Moving Average (EMA)** of the model weights was maintained during training. The EMA model acts as a temporal ensemble of historical network parameters, which generally yields better generalization on unseen data compared to the final epoch weights.

Optimization Strategy

Training was performed using the **Adam optimizer** with weight decay to reduce parameter overfitting. A **OneCycle learning rate schedule** was used to gradually increase the learning rate during the early stages of training and then anneal it towards a lower value, allowing the model to converge to a flatter and more generalizable solution. Gradient clipping was also applied to stabilize optimization and prevent gradient explosion.

Final Training Configuration: In accordance with the Phase-2 assignment instructions, the **training and validation datasets were concatenated into a single dataset containing 18,466 images**. The model was trained for multiple epochs using this combined dataset to maximize the amount of training information available. The EMA-averaged model parameters were saved as the final submission file (`b23cm1003.pth`) so that the model can be directly loaded and evaluated using the provided evaluation script. These improvements emphasize **strong regularization, stable optimization, and enhanced data augmentation**, enabling the ResNet-18 model to achieve improved generalization while maintaining full compliance with the assignment constraints.